

“When purchasing a kit, make sure that it is at the right difficulty level for you. Also consider the number of sensors and number of building blocks in the kit.”—Sachitanand Malewar, Director, NEX Robotics



package deals. For example, instead of offering a bot with ten different kinds of sensors, they could offer a bot with four sensors and the rest six can be bought individually as add-ons.

Displays, chips, networking, etc. Many of these kits use the technology that is currently being used in industries. Robotic kits for engineering students, in particular, are equipped with ultrasonic/infrared sensors, LCDs, latest microcontrollers, etc. Some of these technologies don't even find a mention in the current academic syllabus. So the students are exposed to such devices and concepts through robotics and their personal projects.

Most of the kits that use microcontrollers can be interfaced with PCs, Ethernet modules, Zigbee wireless modules, sensors, etc.

Why is robotics not much popular in India?

In countries like USA, Japan and Korea, most of the high-paying jobs are from high-tech R&D sectors and require constant innovation. As a developed country, there is a good emphasis on high-quality vocational training and universities have talented vocational teachers. Children are exposed to high-tech toys at a very young age. Students work overtime and use self-earned money for the hobby robotics, because of which they have bigger budget for their robotics projects.

In India, we are experiencing all these things for the first time. Most of the critical electrical and mechanical components and systems, including harmonic drives, servo valves and cylinders, servo motors, controllers, encoders and position resolvers, are currently being imported, making the final product extremely expensive.

Also, Indians feel alienated by the idea of robots replacing human labour. It does make sense that in an abundantly populated country like ours, robotics will not be preferred. But now people have started to understand the other uses of robots where humans need not be replaced. Fields like space exploration and defence can flourish with the use of robotics.

Advice to consumers

All the robotics sets have instructions on how to build some basic robots. The keyword here is 'build,' because you will be building them. So make

“The LEGO Mindstorms NXT is based on the ARM processors. All sensors are PC-compliant and motors are geared servo motors with built-in PID controllers.”—Sudhanshu Sharma, managing director, Techtronics Education



sure that this is what you want. When purchasing a kit, make sure that it is at the right difficulty level for you. Also consider the number of sensors and number of building blocks in the kit.

Specifications play a very important role. Make sure you have the right specifications before you order online (or offline). In particular, pay attention to following features of the kit: Programmability and ease of use, number of sensors and type of sensors used, type of batteries and motors used, applications and functionalities of the robot, number of degrees of freedom in case of robotic arms and walking robots, communication systems and cost.

Look at the safety features too. Many a times, robotic kits are sold with lots of claims that are not true. Robots often have insufficient battery power; the battery becomes unusable after a few days of use. So avoid buying kits that use low-cost lithium-ion/NiMH batteries from grey market. These may not have proper battery protection and come with low-cost battery chargers. In such cases, it is very likely that the battery will explode and cause serious injuries after a few charge-discharge cycles. Transistorised motor drivers too can blow if not handled properly. Buy gear motors, which have black gear box and are more reliable.

Look for scalability in terms of add-ons. Buy a kit that is multifunctional, expandable and reprogrammable so that you can keep developing more and more complex projects. For example, a simple line-follower robot kit or mobile-operated robot kit will be nothing but a toy after some time of

usage. A microcontroller-based robot kit or a programmable walking robot kit, on the other hand, will give good knowledge, offer continuous entertainment and learning, and can be used in making more advanced robots.

Ensure that the kit comes with a well-documented manual or a training programme.

An ideal hobby kit would be a microcontroller development board with schematic, standard sensors, reliable wireless modules, DC motors with encoders and a USB interface. The development board helps to interface many different kinds of peripherals that you can develop yourself. ●

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